

IMDb Movie Rating Scraper

PROJECT DESCRIPTION:

The **IMDb Movie Rating Scraper** is a Python-based automation tool that dynamically scrapes movie data from IMDb using Selenium and Chrome WebDriver. It retrieves movie details like title, release year, IMDb rating, and ranking from the **Top 250 Movies** list.

This scraper automates browser actions to handle dynamic content loading. The data extracted can be used for movie trend analysis, building recommendation engines, or for personal film databases.

FEATURES:

1. **Dynamic Movie Scraping**
 - Uses Selenium to load IMDb Top 250 page and extract full content.
2. **Top Movie Rankings**
 - Scrapes movie name, year, and IMDb rating.
3. **Structured Output**
 - Saves extracted data to CSV for easy access and analysis.
4. **Headless Mode (Optional)**
 - Can run without opening a browser window.
5. **Expandable**
 - Easily extendable to scrape individual movie pages for cast, genre, etc.

TECHNOLOGIES USE:

1. **Python:**
 - Language used for scripting.
2. **Libraries:**
 - Selenium: For dynamic page handling.
 - pandas: For data manipulation and export.
 - time: For delay control.
 - webdriver_manager: For auto-downloading ChromeDriver.
3. **Browser Automation:**
 - Chrome with WebDriver for rendering JavaScript-loaded content.
4. **Output Format:**
 - CSV

OUTCOMES:

1. Reliable Data Extraction

- Handles JavaScript-rendered content with Selenium.

2. Up-to-Date Ratings

- Can be scheduled daily to track rating trends.

3. Custom Data Use

- Exported CSV files can be used in data science, dashboards, or ML.